

Loan Approval Prediction & Process Improvement

Functional Requirements Document (FRD)

Build-ready specification for the risk-tiered loan approval workflow

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1. Document Control

Version	Date	Author	Description of Change
0.1	11-07-2024	Chirag Seervi	Initial draft, derived from BRD requirements
1.0	23-07-2024	Chirag Seervi	Finalized functional & non-functional specs

2. Purpose & Traceability

This FRD translates the business requirements defined in the companion Business Requirements Document (BRD) — Loan Approval Risk-Tiered Process — into functional and non-functional specifications a build team can act on. Every functional requirement below traces back to a business objective defined in BRD Section 4.

3. Functional Requirements

ID	Requirement Description	Priority	Linked Business Objective
FR-01	System shall flag an applicant as 'low-risk' when credit history is present AND loan amount is below the low-risk threshold relative to income.	Must	Reduce manual review time for low-risk applicants
FR-02	System shall flag an applicant as 'high-risk' when 2 or more of: no credit history, low income, high loan amount are present.	Must	Reduce default risk from high-risk approvals
FR-03	System shall route low-risk applicants to a streamlined approval path with reduced manual checkpoints.	Must	Reduce manual review time for low-risk applicants
FR-04	System shall route high-risk applicants to an enhanced manual review queue rather than automatic decision.	Must	Reduce default risk from high-risk approvals
FR-05	System shall apply a secondary, smaller positive weighting for graduate status in borderline cases.	Should	Increase decision consistency
FR-06	Dashboard shall display approval/rejection rate broken out by risk tier for ongoing monitoring.	Should	Increase decision consistency
FR-07	System shall log which triage rule(s) triggered for each decision, for audit and compliance review.	Must	Reduce default risk from high-risk approvals

4. Non-Functional Requirements

Category	Requirement
Compliance	Triage logic must exclude protected attributes (gender, location) as direct or proxy scoring inputs, per fair-lending requirements.
Auditability	Every automated triage decision must be logged with the rule(s) that triggered it, retained for [X] years.
Performance	Triage evaluation must complete in under 2 seconds per application to avoid slowing the origination flow.
Usability	Loan officers must be able to see why an application was routed to a given tier, in plain language, in the review UI.

5. User Stories & Acceptance Criteria

Story ID	User Story	Acceptance Criteria
US-01	As a loan officer, I want low-risk applicants automatically routed to a streamlined approval path, so that I can process them faster without redundant manual checks.	Given an applicant with credit history present and a low loan-to-income ratio, when the application is submitted, then it is routed to the streamlined queue and flagged 'low-risk.'
US-02	As a loan officer, I want high-risk applicants flagged for enhanced review, so that I don't approve high-default-risk loans automatically.	Given an applicant with 2+ risk flags (no credit history, low income, high loan amount), when the application is submitted, then it is routed to enhanced manual review and flagged 'high-risk.'
US-03	As a compliance officer, I want every triage decision logged with its triggering rule, so that I can audit for fair-lending compliance.	Given any triaged application, when a decision is logged, then the system records which rule(s) fired and excludes protected attributes from the logic.
US-04	As a risk manager, I want a dashboard view of approvals by risk tier, so that I can monitor whether the new triage process is performing as expected.	Given at least one week of triaged applications, when I open the dashboard, then I see approval/rejection rates broken out by low-risk vs. high-risk tier.

6. Process Flow

Reference: BPMN diagram mapping the as-is single-path review process against the to-be risk-tiered process, showing the decision point where applications split into streamlined vs. enhanced-review paths.

7. UAT Test Plan (Summary)

Test Case ID	Scenario	Expected Result	Linked Requirement
TC-01	Applicant has credit history + low loan amount vs income	Routed to streamlined approval path	FR-01, FR-03
TC-02	Applicant has no credit history + low income + high loan amount	Routed to enhanced manual review	FR-02, FR-04
TC-03	Borderline applicant, graduate status only differentiator	Small positive weighting applied, does not override risk flags	FR-05
TC-04	Any triaged application	Rule trigger logged; no protected attributes present in log	FR-07

8. Open Questions / Parking Lot

- What loan-amount-to-income ratio threshold defines 'low risk' vs. 'high risk'? Needs Risk team sign-off.
- Should the streamlined path still require any human sign-off, or can it be fully automated for the lowest-risk tier?
- How will edge cases (e.g., self-employed applicants with variable income) be handled within the triage logic?

9. Related Documents

- Business Requirements Document (BRD) — Loan Approval Risk-Tiered Process (companion document)
- Source analysis: SQL-based approval/rejection driver analysis (Python cleaning + EDA, SQL business analysis)
- Stakeholder dashboard, BPMN process diagram, Jira backlog